

Git Masterclass

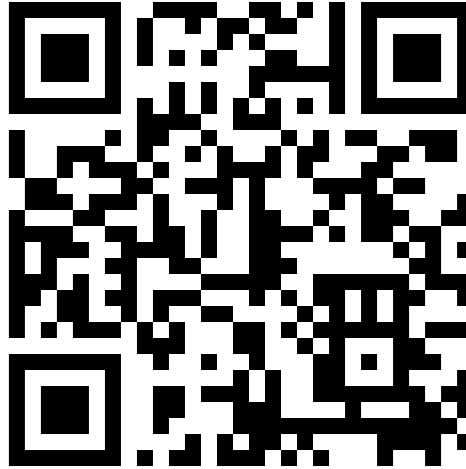
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Setup

Setup

Slides & Notes



URL macconville.ie/masterclass

Downloading Git

Instructions <https://git-scm.com/install>

Linux Use your package manager

Mac Package manager (brew.sh) or Xcode

Windows Git Bash

Setup

Setting Up A Forge

GitHub github.com

GitLab gitlab.com

Codeberg codeberg.org

Forgejo forgejo.org

sourcehut cgit

Connecting GitHub

- SSH! <https://docs.github.com/en/authentication/connecting-to-github-with-ssh>

Getting a GUI

Official list git-scm.com/tools/guis

GitHub Desktop github.com/apps/desktop

Windows Download from link

Mac Brew formula available

Linux codeberg.org/ckruse/Gitte

Editor Integration

- RStudio
- VSCode <https://code.visualstudio.com/docs/sourcecontrol/quickstart>

Motivation

Motivation

Workflow

Demo

Goals

- By the end of this workshop, all your code will be backed up, and be done easily and continually from now on

Version Control

- A way of backing up, and also versioning any text
- A free and fairly easy and quick way to ensure you **don't ever lose your code**
- Also keeps track of versions, no more **FINAL, FINAL (2), LAST EDIT, COPYv2**
- Share and collaborate more easily
- Reproducibility
- Keep everything organised in one place

What to Version Control

- Everything text
 - Code
 - Papers (if not already on Overleaf)
 - Notes
 - Configuration files (dotfiles)

Datasets?

- A small to mid size csv maybe
- Possibly not pure text, so not such a good fit for VCS like git
- Maybe too big
- But some will be appropriate
- Should be backed up *somewhere!*

Reproducibility

- Personal
 - I want you to ask yourself, how long would it take to restore all your work on a brand new machine, and how difficult/easy would it be?
 - No need for hypothetical, let's test right now!
- For other researchers / users
- Licence!
- Demo

Other Things

- Testing
- CI/CD
- Releases

Git

- Most popular and thus well supported and resourced version control system

GitHub

- Most popular and thus well supported and resourced git forge
- Not official!

Fun Things

- GitHub pages site
- Social network features
 - Stars
 - Following
- cinema!

Introduction

Links Again



URL macconville.ie/masterclass

Resources

- <https://wizzardzines.com/git-cheat-sheet.pdf>
- <https://ohshitgit.com>
- <https://git-scm.com/book/en/v2>
- <https://wizzardzines.com/zines/git>
- <https://book.the-turing-way.org/reproducible-research/vcs>

Model

- Distributed version control system
- It's just files in `.git/`
- Graphs!
- Hashs

Setting up a repo

```
git init
```


Status

```
git status
```

Remotes

`git` remote

Adding

```
git add
```

Committing

- `git commit`
- `git commit --amend`

Ignoring

- `.gitignore`
- `.DS_Store`
- `__pycache__`
- `.Renviron`
- Any other examples

Seeing History

```
git log
```

Branching

- `git` branch name
- `git` switch name
- Create with -c flag

Diffing

- `git diff`
- `git diff --staged`

Pushing

```
git push
```

Pulling

- `git pull`
- Not to be confused with a pull request

Cloning

```
git clone url
```

Configuring

- `git config --list --show-origin`
- `git config --global user.name "Your Name"`

Merging

- `git` merge name
- Conflicts

Tagging

- `git tag name`
- `git push --tags`

Throwing Away Changes

- `git restore name`

Undoing Change

- `git revert hash`
- `git checkout hash -- path/to/file`

The Nuclear Option

`rm -rf`

Collaborating!

- Add other people to private repos
- Or just leave it public
- Find a project

Introduction

Hooks

Actions

<https://docs.github.com/en/actions>

Exercise

You should all o